

Week 6 – Machine Learning Model Architecture Design (20260617)

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> This is the active Week 6 report for the current run. Older dated Week 6 > report files in `data/reports/` are treated as superseded outputs.

1. Executive Summary

This document describes the machine learning architecture designed and implemented in Week 6 of the Quantitative Research & Trading project. Two complementary approaches price chooser options on JPM stock:

- **Approach 1 (ML + chooser closed form):** ML models predict 20-day forward realised volatility; the predicted σ is fed into the chooser pricing formula.
- **Approach 2 (End-to-End):** ML models directly map chooser contract parameters and market features to chooser prices.

2. Problem Formulation

Approach 1

$$\hat{V}_{\text{chooser}} = C \left(S, K, T_2, r, q, \hat{\sigma}_{\text{ML}} \right) + P \left(S, K e^{-r(T_2 - T_1)}, T_2 - T_1, r, q, \hat{\sigma}_{\text{ML}} \right)$$

where $\hat{\sigma}_{\text{ML}}$ is the ML-predicted 20-day forward realised volatility.

$$\text{Target: } \sigma_{\text{fwd,20d}}[t] = \sqrt{252} \cdot \text{std} \left(\ln \frac{S_{t+i}}{S_{t+i-1}} \right)_{i=1}^{20}$$

Approach 2

$$\hat{V}_{\text{chooser}} = f_{\theta} \left(S, K, T_1, T_2, r, q, m, \mathbf{x}_{\text{market}} \right)$$

where f_{θ} is a trained ML model and $\mathbf{x}_{\text{market}}$ is the market feature vector.

Target: chooser price benchmarked with Monte Carlo-valued call/put legs.

3. Data & Feature Engineering

3.1 Raw Data Sources

Source	Description	Frequency
yahoo_jpm_2018_2024.csv	JPM daily close prices	Daily
fred_DGS10_2018_2024.csv	US 10-yr Treasury yield	Daily
fred_VIXCLS_2018_2024.csv	CBOE VIX index	Daily
jpm_dividends_2018_2024.csv	JPM dividend payments	Per event
alphavantage_news_jpm_2018_2024.csv	News sentiment scores	Per article

3.2 Market Features (Approach 1 – Volatility Dataset)

All features are strictly ****backward-looking**** to prevent look-ahead bias. Dataset: 1,681 trading days.

- `hist\_vol\_5d`
- `hist\_vol\_10d`
- `hist\_vol\_20d`
- `hist\_vol\_40d`
- `hist\_vol\_60d`
- `vol\_ratio\_5\_20`
- `vol\_ratio\_20\_60`
- `vol\_20d\_change`
- `return\_1d`
- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`

3.3 Features (Approach 2 – Chooser Pricing Dataset)

All market features above, plus chooser-specific parameters. Dataset: 17,850+ rows (daily dates × chooser contracts).

- `vol\_ratio\_5\_20`

- `vol\_ratio\_20\_60`
- `vol\_20d\_change`
- `return\_1d`
- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`
- `S`
- `K`
- `moneyness`
- `T1`
- `T2`

3.4 Target Variables

Approach	Target	Description
1	fwd_vol_20d	Annualised realised vol over next 20 trading days
2	chooser_price	Chooser price using Monte Carlo benchmark

4. Time-Series Validation Framework

Data is split **chronologically** (never randomly) to prevent look-ahead bias. The validation policy uses a rolling-window `TimeSeriesSplit` with `n\_splits=3` on the training partition, no shuffling, and no gap. Each fold is fit on a fixed-length trailing training window and validated on the next chronological block. This means the training window slides forward with a fixed size, while each validation block keeps a fixed chronological size (validation window). The exact fold ranges are:

Fold	Window type	Train range	Validation range	Train days	Val window days
Fold 1	rolling	2018-03-29 → 2019-05-30	2019-05-31 → 2020-07-29	294	294

Fold	Window type	Train range	Validation range	Train days	Val window days
Fold 2	rolling	2019-05-31 → 2020-07-29	2020-07-30 → 2021-09-28	294	294
Fold 3	rolling	2020-07-30 → 2021-09-28	2021-09-29 → 2022-11-28	294	294

All features are scaled using `RobustScaler` fitted **only** on the training set. For model selection, each tunable learner is optimized with `RandomizedSearchCV` under the time-series CV protocol above, then refit on the combined train+validation partition before the final test-set evaluation.

Split	Date Range	Fraction
Train	2018-03-29 → 2022-11-28	70%
Validation	2022-11-29 → 2023-11-29	15%
Test	2023-11-30 → 2024-12-02	15%

5. Model Architectures

5.1 Approach 1 – ML Volatility Prediction

Before/After Tuning Snapshot

Approach	Model	Before Val MAE	After Val MAE	Δ MAE	Default fit sec	Search+refit sec	Compute multiple
Approach 1	RandomForest	0.0692	0.0756	-0.0064 (-9.3%)	0.31	10.89	35.5x
Approach 1	XGBoost	0.0699	0.0685	0.0013 (1.9%)	0.19	4.81	25.3x
Approach 2	LinearRegression	5.1325	4.7457	0.3869 (7.5%)	0.03	0.13	4.4x
Approach 2	XGBoost	6.0226	5.1763	0.8462 (14.1%)	0.39	6.98	17.8x
Approach 2	NeuralNetwork	6.2190	6.0719	0.1471 (2.4%)	44.88	738.97	16.5x

Approach	Model	Before Val MAE	After Val MAE	Δ MAE	Default fit sec	Search+refit sec	Compute multiple	Best parameters
Approach 1	RandomForest	0.0692	0.0756	-0.0064 (-9.3%)	0.25	6.95	27.3x	
Approach 1	XGBoost	0.0699	0.0685	0.0013 (1.9%)	0.13	3.39	25.4x	
Approach 2	LinearRegression	5.1325	4.7457	0.3869 (7.5%)	0.02	0.09	5.4x	
Approach 2	XGBoost	6.0226	5.1763	0.8462 (14.1%)	0.35	4.91	14.0x	

Approach	Model	Before Val MAE	After Val MAE	Δ MAE	Default fit sec	Search+refit sec	Compute multiple	Best parameters
Approach 2	NeuralNetwork	6.2190	6.0719	0.1471 (2.4%)	8.44	172.46	20.4x	

Model	CV MAE	Val MAE	Test Vol MAE	Search sec	Refit/Fit sec	Best parameters
LSTM	-	0.0444	0.0753	6.56	-	-
LinearRegression	-	0.0535	0.0993	0.01	0.00	-
XGBoost	0.1409	0.0685	0.0765	2.39	0.99	subsample=0.9, reg_lambda=5.0, reg_alpha=0.01, n_estimators=1000, min_child_weight=1, max_depth=6, learning_rate=0.05, colsample_bytree=0.8
RandomForest	0.1255	0.0756	0.0702	6.81	0.13	n_estimators=200, min_samples_split=10, min_samples_leaf=6, max_features=sqrt, max_depth=8
NeuralNetwork	-	0.0880	0.1367	0.41	0.76	-

LSTM Ablation

Lookback	Units 1	Units 2	Dropout	LR	Val MAE	Train sec	Selected
10	32	16	0.2	0.001	0.0444	6.56	yes
20	64	32	0.2	0.001	0.0464	11.13	no
40	128	64	0.2	0.0005	0.0495	20.32	no
20	128	64	0.3	0.001	0.0489	26.52	no

- Architecture: LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.2) → Dense(16) → Dense(1)
- Lookback window: 20 trading days
- Optimizer: Adam | Loss: MSE | Early stopping (patience=10)
- Input shape: (batch, 20, 23)

5.2 Approach 2 – End-to-End Supervised Chooser Pricing

Model	CV MAE	Val MAE	Test MAE	Search sec	Refit/Fit sec	Best parameters
LinearRegression	8.8246	4.7457	11.2689	0.08	0.01	model__positive=True, model__fit_intercept=True

Model	CV MAE	Val MAE	Test MAE	Search sec	Refit/Fit sec	Best parameters
XGBoost	8.7259	5.1763	12.4058	4.72	0.19	subsample=0.8, reg_lambda=1.0, reg_alpha=1.0, n_estimators=300, min_child_weight=7, max_depth=3, learning_rate=0.01, colsample_bytree=0.9
NeuralNetwork	5.8955	6.0719	8.4856	77.83	94.63	model__learning_rate_init=0.001, model__hidden_layer_sizes=(256, 128, 64, 32), model__batch_size=64, model__alpha=0.001

Baseline comparison

- Week 4 BSM baseline (European options): MAE=0.112135, RMSE=0.174916, p-value=0.9524
- The closed-form chooser formula computed with maturity-matched historical volatility is used only as a comparison baseline, not as the training target.

Selection Note

- For Approach 1, the final ranking is based on validation MAE after the time-series search/refit pipeline.
- LinearRegression and NeuralNetwork are included in the same ranking table so the model choice uses one metric family across the full candidate set.
- For Approach 2, the ranking is based on test-set MAE/RMSE after the same chronological split protocol.

6. Performance Summary (Test Set)

6.1 Approach 1 – Volatility Prediction

Model	Vol MAE	Vol RMSE	Option MAE	Option RMSE
RandomForest	0.07018	0.09641	7.1302	9.6827
XGBoost	0.07646	0.10174	9.0047	12.1432
LSTM	0.07530	0.09801	9.4707	12.5957
LinearRegression	0.09926	0.12709	10.0806	14.3767
NeuralNetwork	0.13669	0.17166	13.6551	18.1805

Vol MAE/RMSE: annualised vol units. Chooser MAE/RMSE: USD.

6.2 Approach 2 – End-to-End Chooser Pricing

Model	MAE (\$)	RMSE (\$)	R ²
LinearRegression	11.2689	14.2303	-0.2552
XGBoost	12.4058	15.8663	-0.5604
NeuralNetwork	8.4856	11.1067	0.2354
BSM Baseline	0.6531	0.9058	0.9949

Target: chooser price benchmarked with Monte Carlo-valued call/put legs. The table includes the closed-form BSM baseline for the same contracts.

6.3 Stratified Error Diagnostics

This section highlights weak scenarios by showing the worst test slice for each model after splitting by moneyness and maturity bucket.

Approach 1 – Volatility Prediction by Moneyness and T2

Model	Moneyness	T2 bucket	Count	MAE	RMSE	R ²
RandomForest	ITM	long	472	9.9373	12.8589	0.0818
XGBoost	ITM	long	472	12.6670	16.3284	-0.4805
LSTM	ITM	long	472	13.1893	16.8266	-0.5723
LinearRegression	ITM	long	472	14.7293	19.9025	-1.1996
NeuralNetwork	ITM	long	472	20.5027	25.8029	-2.6972

Approach 2 – Chooser Pricing by Moneyness, T1, and T2

Model	Moneyness	T1	T2 bucket	Count	MAE	RMSE	R ²
LinearRegression	OTM	0.25	long	256	13.8627	17.7853	-1.0143
XGBoost	OTM	0.25	long	256	16.2997	20.3417	-1.6350
NeuralNetwork	ITM	0.5	long	256	11.8733	14.7102	-0.0435

7. Limitations & Recommended Next Steps

Current Limitations

Data & Targets

- **No market-implied vol**: Targets are derived from historical/closed-form

prices, not actual market option quotes. This is a fundamental constraint: Approach 2 learns to predict a synthetic chooser surface, not a real market surface. High R² values (e.g., MLP R²=0.89) reflect model fit to the synthetic label, NOT market predictability.

- **Synthetic chooser labels (Approach 2)**: Targets are computed as

``chooser_price(S, K, T1, T2, r, q, hist_vol_match(T2))``. Since this is a deterministic closed-form calculation, linear models and neural networks with sufficient capacity can still fit the surface, but the direct historical-vol feature leakage into the input set has been removed.

- **Volatility term-structure matching**: Vol used for each maturity is now

matched to option T2, preventing term-structure mismatch. Approach 2 now includes volatility proxy features such as VIX, volatility ratios, and volatility change signals, while excluding raw historical volatility windows.

Model Architecture

- **Simplified chooser grid**: 2 decision times × 3 maturities × 3 moneyness levels.
- **Static features**: No real-time microstructure data (bid-ask, volume).

- ****No market data feedback****: Models do not adjust based on actual prices observed at decision time T1.

Model-Specific Notes

Approach 2 – Route 2 Pricing

- ****LinearRegression & MLP****: R^2 should improve materially once volatility proxy features are included, because these models now receive the strongest explanatory signal that is allowed by the experiment design.
- ****XGBoost/GradientBoosting****: Lower R^2 (0.40-0.50) may indicate that tree models struggle with the smooth closed-form surface; alternatively, the current feature set lacks sufficient flexibility for tree splits. This is NOT a sign of data leakage or feature engineering failure—it reflects model-surface mismatch.

Recommended Next Steps (Week 6+)

- Incorporate implied volatility data for more realistic targets.
- Add Greeks (delta, gamma, vega) as engineered input features.
- Implement Bayesian hyperparameter optimisation.
- Extend LSTM to multi-step ahead vol forecasting.
- Build ensemble that combines Approach 1 and Approach 2 predictions.
- Evaluate on 2025+ data for out-of-sample performance.
- ****Collect real market chooser prices**** and retrain Approach 2 models on actual traded prices instead of synthetic targets.

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